



Muhammad Talal Khan

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ABOUT ME

Computer Engineering graduate experienced in neural networks, federated learning, neuromorphic computing, and quantum technologies, with hands-on experience in designing and implementing projects from concept to execution. Committed to learning and applying emerging technologies to solve real-world problems.

EDUCATION AND TRAINING

2022 – 2026 Abbottabad, Pakistan

BCE (BACHELOR IN COMPUTER ENGINEERING) COMSATS University Islamabad, Abbottabad Campus

Website <https://cuiatd.edu.pk/> | **Final grade** CGPA 3.67/4.00

PROFESSIONAL EXPERIENCE

06/2026 – CURRENT

Engineering Intern at IEON, UK

Working remotely on an AI-based predictive maintenance project for industrial machinery, using Eulerian Video Magnification (EVM) to extract subtle vibration patterns from video for early fault detection.

03/2025 – 08/2026

IEEE Student Branch, Vice President

COMSATS University Islamabad, Abbottabad Campus

Served as Vice President of the IEEE Student Branch, supporting chapter activities, coordinating events, and contributing to student engagement and professional development.

HONOURS AND AWARDS

11/08/2026

Best Final Year Design Project Award | InnovatEx 2026 – COMSATS University Islamabad, Abbottabad Campus

Awarded Best Final Year Design Project at InnovatEx 2026.

Link <https://lnkd.in/p/d62Vwu49>

Campus Honor Roll Award – COMSATS University Islamabad, Abbottabad Campus

Recipient of the Campus Honor Roll Award Thrice for academic excellence; official recognition video featured on the university's official platform (link below).

Link <https://www.facebook.com/share/v/1F4HMkQiuJ/>

2023

Prime Minister Youth Laptop Scheme Award – Higher Education Commission (Pakistan)

Awarded a laptop under the Prime Minister Youth Laptop Scheme 2023 for merit and achievement.

● **PROJECTS**

CURRENT
Experimenting Quantum Augmentations with Cutting Edge Classical Technologies

Final Year Project: implementing latest classical technologies (like Federated Learning, Neuromorphic Computing) on advanced platforms (like modern Microcontrollers and SBCs, FPGAs, APSoCs) and experimenting augmentations of Quantum Technologies with them to try and achieve further optimizations.

Neural Network-Based American Sign Language Recognition

Developed a neural network-based ASL gesture recognition system using image preprocessing, feature extraction, and classification to translate hand gestures into meaningful predictions.

Implementation of Machine Learning Models Using TinyML on ESP32-S3

Implemented TinyML models on ESP32-S3, enabling on-device real-time data processing and intelligent decision-making on resource-constrained hardware.

CURRENT
Physics-Based Heat Diffusion Simulation and Synthetic Data Generation

Developed a physics-based 3D heat diffusion simulator using the Finite Difference Method (FDM) to simulate temperature fields for synthetic data generation and ML model training.

Clean vs. Noisy Image Classifier using Machine Learning

Developed a binary image classification system to distinguish clean and noisy images. The project involved dataset loading, feature extraction, preprocessing, and model training, focusing on efficient and accurate classification.

Autonomous Robotic Car with Sensor-Based Navigation

Developed an automated robotic car using microcontrollers and sensors for real-time obstacle detection and navigation, enabling self-directed movement and intelligent path planning.

● **SKILLS**

C/C++ | Python | VHDL - Hardware Description Language | Embedded Linux (Petalinux, Buildroot, Yocto), SoC (FPGA) | Field Research

● **HOBBIES AND INTERESTS**

Gymnastics and Marshal Arts

Engaged in gymnastics and martial arts training, demonstrating discipline, physical conditioning, and commitment to continuous improvement, with personal performance shown in the link below.

Links <https://vt.tiktok.com/ZSa2FdkpV/> | <https://vt.tiktok.com/ZSa2YY6qH/>

Book Reading

● **LANGUAGE SKILLS**

Mother tongue(s): **HINDKO**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
URDU	C2	C2	C2	C2	C2